

IKEY, Russian Federation

WMO: 305070

Lat: 54.183N

Lon: 100.083E

Elev: 527

StdP: 95.15

Time Zone: 8.00 (E08)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.8	-34.4	-39.6	0.1	-37.8	-36.1	0.1	-34.3	9.7	-13.7	5.4	-15.4	0.2	320	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	27.0	17.9	25.3	17.3	23.6	16.7	19.6	24.7	18.5	23.6	17.4	22.3	2.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.4	22.7	16.4	12.4	21.4	15.4	11.7	20.1	58.4	24.8	54.6	23.7	51.3	22.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	5.5	4.3	-41.0	30.4	2.9	2.1	-43.1	31.9	-44.8	33.1	-46.4	34.3	-48.5	35.9
			-41.2	21.0	2.7	1.1	-43.2	21.8	-44.8	22.5	-46.4	23.1	-48.4	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.0	-20.6	-17.5	-9.1	0.5	8.4	13.6	16.6	14.6	8.0	0.5	-10.4	-18.2
	DBStd	14.02	6.81	6.36	5.89	4.50	4.26	3.62	2.95	2.74	3.58	4.31	6.67	6.92
	HDD10.0	4550	949	769	593	284	82	10	1	2	79	296	612	873
	HDD18.3	7092	1207	1002	851	534	307	146	67	119	312	554	862	1132
	CDD10.0	518	0	0	0	0	33	118	206	143	18	0	0	0
	CDD18.3	20	0	0	0	0	0	4	14	2	0	0	0	0
	CDH23.3	409	0	0	0	0	42	121	198	45	3	0	0	0
	CDH26.7	68	0	0	0	0	7	23	33	5	0	0	0	0
(14) Wind	WSAvg	1.3	0.8	1.0	1.4	2.1	2.0	1.7	1.4	1.2	1.3	1.2	1.1	0.8
(15) Precipitation	PrecAvg	432	12	9	10	18	36	67	85	85	52	24	19	15
	PrecMax	645	22	19	24	47	69	172	162	161	108	60	40	29
	PrecMin	273	3	4	2	6	18	28	25	28	17	5	9	6
	PrecStd	84	5	4	6	9	13	33	30	27	22	12	9	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	2.6	9.9	17.4	27.6	29.2	29.7	26.7	23.4	17.4	7.4	1.3
		MCWB	-2.9	-0.5	4.6	8.5	14.8	17.7	20.2	18.8	14.9	10.0	3.5	-1.6
	2%	DB	-5.9	-2.2	5.9	14.0	24.0	26.6	27.5	24.2	20.2	12.8	2.7	-3.2
		MCWB	-7.3	-4.5	1.5	6.8	13.2	16.6	19.3	17.4	14.2	7.7	-0.1	-4.8
	5%	DB	-8.7	-5.0	3.3	11.3	21.3	24.7	25.6	22.9	17.4	9.9	0.2	-6.0
		MCWB	-9.6	-6.9	-0.3	5.4	11.6	16.5	18.4	16.8	12.5	5.9	-1.8	-7.3
	10%	DB	-11.2	-7.6	1.0	8.4	18.6	22.1	23.9	21.3	14.8	7.3	-1.7	-8.1
		MCWB	-11.7	-8.9	-1.7	3.5	10.4	15.0	18.0	16.2	11.0	4.2	-3.2	-9.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-0.1	5.0	9.4	15.7	19.5	21.8	20.2	16.4	10.9	3.4	-1.7
		MCDB	-1.3	2.9	9.5	15.7	25.7	25.2	27.8	24.7	21.2	16.2	7.2	0.8
	2%	WB	-7.2	-4.4	1.8	7.3	13.8	18.0	20.2	18.5	14.3	8.1	0.2	-4.8
		MCDB	-6.2	-2.3	5.6	13.7	22.7	25.0	25.6	22.8	19.2	12.2	2.5	-3.3
	5%	WB	-9.7	-6.7	-0.1	5.8	12.4	16.8	19.2	17.3	13.0	6.2	-1.8	-7.0
		MCDB	-8.9	-5.0	3.0	11.0	19.9	23.3	24.5	21.7	16.4	9.2	-0.2	-6.0
	10%	WB	-11.8	-8.9	-1.8	3.8	10.8	15.6	18.2	16.4	11.4	4.5	-3.1	-8.9
		MCDB	-11.2	-7.6	1.0	8.0	17.4	21.1	23.1	20.5	14.7	7.1	-1.7	-8.1
(35) Mean Daily Temperature Range	MDBR		11.3	14.6	14.7	11.2	13.8	12.3	11.3	10.8	11.4	10.4	10.5	10.6
	5% DB	MCDBR	12.4	16.7	17.3	16.6	20.8	18.2	15.3	15.2	17.1	15.8	11.2	12.8
		MCWBR	11.2	14.4	13.4	10.7	11.8	9.5	8.4	8.9	10.7	11.0	8.8	11.1
	5% WB	MCDBR	12.1	16.3	16.6	15.7	18.9	15.1	13.6	13.6	15.4	14.9	10.5	12.2
		MCWBR	11.0	14.2	12.8	10.4	11.2	8.6	8.2	8.4	10.2	10.6	8.5	10.7
(40) Clear-Sky Solar Irradiance	taub		0.271	0.304	0.343	0.381	0.421	0.417	0.417	0.390	0.345	0.325	0.300	0.274
	taud		2.087	2.064	2.084	2.204	2.138	2.258	2.291	2.345	2.423	2.354	2.181	2.047
	Ebn at Noon		658	762	818	842	824	833	825	826	824	749	619	555
	Edh at Noon		68	99	123	126	142	128	121	108	88	75	62	58
(44) All-Sky Solar Radiation	RadAvg		1.03	2.05	3.58	4.72	5.23	5.78	5.24	4.47	3.19	1.89	1.01	0.70
	RadStd		0.07	0.12	0.18	0.29	0.45	0.34	0.46	0.31	0.29	0.14	0.09	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page